A Smart Computing Framework for AI-Driven Urban Drainage and Waterlogging Prediction

Sonam Singh[1], Tushar Chavhan[2], Sakshi Ingale[3], Vrushali Dhage[4], Snehal Godage[5], Shubhangi Bhawari[6]
MIT Academy of Engineering, Alandi, Pune, Maharashtra, India[1]
Dr. D. Y. Patil Institute of Technology, Pimpri, Pune, Maharashtra, India[2,3,4,5,6]
[1]sonamchauhan346@gmail.com, [2]chavhantushar223@gmail.com, [3]sakshiingale2212@gmail.com
[4]vrushalidhage19@gmail.com, [5]sgodage46@gmail.com, [6]shubhangi.bhawari@dypvp.edu.in

***Abstract*—A multi-source Artificial Intelligence (AI) framework for enhancing and maintaining urban waterlogging prediction is presented in this paper. It achieves this through the combination of actual weather data, drainage maps created using a Geographic Information System (GIS), and Internet of Things (IoT) water level sensors, providing highly accurate and current flood information. The increasing issues of urban flooding, made worse by climate change, rapid urban growth, and poor drainage systems, are the main reasons for this study. To better understand how drainage works, the proposed framework combines information from IoT-based water level networks, spatial drainage models, and data from local weather stations. Using AI methods such as ensemble learning and Long Short-Term Memory (LSTM) neural networks, the system analyzes both live and historical data to make reliable and data-driven predictions of incidents of possible waterlogging. Integration of several data sources, dynamic updating of risk profiles through real-time sensor feedback, and the combination of data-driven AI models and hydrodynamic simulations are what make it novel. The framework's effectiveness is demonstrated through structured simulation-based experimental validation. Overall, this framework advances the development of intelligent and climate-resilient urban drainage systems, supporting the more general objectives of adaptive and sustainable urban development.**

***Index Terms*—GIS-based drainage mapping, IoT water-level sensing, LSTM neural networks, ensemble learning, real-time monitoring, climate-resilient infrastructure, smart cities, artificial intelligence (AI), multi-source data fusion, sustainable drainage systems, waterlogging prediction, urban flood management.**

I. Introduction

Due in large part to shifting rainfall patterns, fast urbanization, and antiquated drainage systems, urban waterlogging has become an increasingly widespread and enduring problem in cities all over the world [1]. With a large percentage of paved and impermeable surfaces, modern cities are particularly vulnerable; even light rainfall can now result in flooded streets, traffic jams, financial losses, and health hazards for the general public [2]. As climate change brings more heavy and unpredictable rainfall, events that were once thought to be extreme are happening more often [1]. Flooding is exacerbated in many urban areas through overburdened drainage systems, dwindling natural water bodies, and improper waste management.

Despite increasing use of artificial intelligence to forecast flooding, most of the existing systems are inefficient and ineffective due to their heavy reliance on either simulations or a single data source [3]. Although hydrodynamic models are precise, they require substantial computational resources and are often unsuitable for real-time decision-making in busy metropolitan regions [4]. In the same way, AI models that depend exclusively on Meteorological data and flood records frequently have information deficiencies, a deficiency in spatial information, and an incapacity to adapt to novel or severe climatic conditions [5]. The failure of multiple systems to effectively represent real-world conditions results in the inability to consider the complexity of how water moves through drainage systems or integrate real time information from IoT sensors [6].

The limitations of these systems result in reduced effectiveness in managing flooding and can reduce the overall long-term sustainability of urban areas [7]. Low-income communities with high levels of vulnerability are often disproportionately affected by waterlogged homes, contaminated drinking water and increased rates of disease due to flooding [1]. A multi-faceted approach that incorporates a comprehensive real-time system utilizing real-time water level monitoring using IoT technology along with GIS drainage mapping utilizing current weather data will improve forecast accuracy and allow for better decision making for emergency responders and urban planners to design more adaptive, sustainable, and resilient urban environments [8] [9].

II. Literature Review

A great deal has been achieved in predicting and detecting urban flooding and obstruction in urban drainage systems due to recent improvements in Artificial Intelligence (AI) and Machine Learning (ML) technology [4]. Predictive models such as Random Forests, Long Short-Term Memory (LSTM) Networks, and Ensemble Techniques are currently capable of generating fast and accurate predictions regarding water levels and the likelihood of a flood [7]. Compared with traditional hydraulic simulation-based approaches that often require large amounts of processing time and fixed data sets, these modern AI-based predictive models can quickly adapt to the ever-changing urban environment and its complexities [10]. Additionally, the combination of AI and physical simulation enables near-real-time forecasting of urban floods; allowing for proactive planning of maintenance activities and emergency responses [2]. However, while modern predictive models have many advantages, their predictive accuracy is highly dependent upon the quality, variability, and consistency of the input data [3].

The Internet of Things (IoT) has also created new possibilities for the continuous monitoring and management of Urban Drainage Systems, enabling city managers and engineers with real-time data to make informed decisions [6]. IoT devices that are networked together can monitor water levels, flow rates, debris accumulation, and automatically control pumps and valves to prevent an overflow in case of a storm event. Using Satellites, the City will have the ability to predict heavy rainfall events with unprecedented detail and lead times, and anticipate potential blockages, thus providing early flood warnings when combined with AI Analytics [5]. Applications in the real world have demonstrated that this combination can increase the cost-effectiveness and proactiveness of drainage management [9]. However, the majority of these implementations are still small-scale, with problems like erratic connectivity and challenges when trying to scale to larger, older, and more varied city infrastructures.

In order to comprehend how water flows through cities, Geographic Information Systems (GIS) are also essential [1]. GIS tools are useful for mapping drainage networks, identifying areas that are vulnerable to flooding, and analyzing the interactions between infrastructure, terrain, and land use [11]. GIS enhances spatial accuracy and assists in identifying potential waterlogging locations and times when combined with AI [12]. However, it is still rare that the full integration of GIS with AI analytics and real-time IoT data, which narrows down the possibility of truly comprehensive flood forecasting systems [10].

A multidisciplinary research approach that integrated different data types was used, such as meteorological data, hydrological patterns, spatial layouts, and real-time sensor readings—into unified systems in order to get around these constraints [7]. This model leverages multiple sources of data, compared to models that use one data source. The fusion of source data enables the making of more accurate, real-time, and context-aware predictions [4]. However, there are still issues in integrating these different data streams in a smooth manner, ensuring the quality of the data, and maintaining interoperability between systems operated by different organizations or built with various technologies.

Despite advancements, a large number of AI-based drainage solutions still depend on discrete datasets, which restrain their ability to adapt and survive in operational situations [13]. There are very few frameworks that successfully combine IoT sensor feedback, GIS-based drainage mapping, and meteorological data onto one single, coherent platform [8]. Scaling and integration are difficult since there are fragmented data infrastructures and a lack of standardized protocols to exchange data [3]. Furthermore, long-term resilience and sustainability aspects such as infrastructure decay, the impacts of climate change, and the need for equitable access to flood protection—must be frequently reconsidered [7]. These needs must be addressed in order to develop urban drainage systems that are both adaptable and intelligent as they provide a foundation for sustainable and resilient urban growth within modern cities.

III. Related Work

TABLE I
COMPARISON OF AI AND MACHINE LEARNING METHODS IN URBAN DRAINAGE SYSTEMS

Approach	**Strengths**	**Limitations**	**Typical Applications**
Supervised Learning	When trained on historical rainfall and drainage data, it can accurately predict floods.	It also relies a lot on labeled datasets and may not be able to adapt quickly to changing conditions.	Mapping the risk of flooding and predicting runoff.
Unsupervised Learning	Finds hidden trends and connections in data that doesn't have labels. This is helpful for finding patterns that aren't normal.	Less useful for tasks that require accurate forecasting because it doesn't predict well.	Grouping land uses and finding strange patterns in how water drains.
Deep Learning	Models complex patterns in time and space well enough to make reliable predictions about water levels.	Needs a lot of data and processing power; hard to understand.	Monitoring water levels and predicting floods in real time.
Reinforcement Learning	Learns how to make real-time control actions better by constantly adjusting to changes in the system.	Performance depends on well-designed reward functions and a lot of tuning.	Smart control of pumps and valves to improve drainage.
Hybrid Approaches	Uses a mix of AI techniques to make the system more accurate, flexible, and resilient overall.	Model integration can make design and interpretation more difficult.	Full flood prediction and automatic drainage management.

IV. Proposed Methodology

AI for Predicting Waterlogging and Drainage

1. Source of Input Data

To fully understand urban drainage systems, the framework will utilize a variety of data from multiple sources including:

- **Real-time meteorological data:** Real-time meteorological data (meteorological stations and forecasting services) that provide current information about the rain's attributes (duration, intensity, etc.) to support effective flood prediction and analysis [1].
- **GIS-based maps:** High resolution geographic information system (GIS) maps illustrate the network of components such as pipes, catch basins, manholes and their spatial relationships with other infrastructure component such as roads and terrain features [11].
- **IoT water level sensors:** Smart sensors allow continuous monitoring of water levels, flow rates and irregular

fluctuations that assist in detecting issues such as blockages or saturated soils which enable timely responses to environmental changes [6].

2. Architecture of AI Models

The system is based on a deep-learning model which learns the temporal patterns of rainfall as well as the interaction between the special characteristics of urban geography and the temporal changes in the rainfall patterns.

- **Time-Based Patterns:** The model studies the rainfall and water level data by using the time-based patterns of the data with special neural network layers such as BiTCN and GRU to understand both the short-term and long-term trends in the data [8].
- **Space-Based Patterns:** The model also uses GIS data to study the structure, capacity and connection of the drainage network of the area to learn about how the geographical configuration of the area influences the movement of the water and where the water will collect [12].
- **Module Integration:** By combining the time-based module with the space-based module, the system generates very accurate flood predictions that are conditioned on the different conditions in different parts of the city [10].

3. Data Fusion Strategy

A framework brings together data from many sources, including sensor readings, historical rainfall data and drainage maps, to provide better flood forecasting models than those of a simple model that is based only on one type of data source.

- **Feature-Level Fusion:** The integration of these data will enable an enhanced understanding of how the varied factors contribute to flooding in specific areas at specific times; and, provide a much larger perspective of the flood process [9].
- **Hybrid Ensemble Learning:** This method incorporates both established hydrologic models with data-driven AI approaches to improve the reliability of the flood predictions by comparing the results of the hybrid ensemble model to existing simulations [3].
- **Dynamic Model Updates:** The model will be able to adjust to changes in weather patterns and/or infrastructure developments over time by incorporating new data into its processes; thereby enabling the model to continue providing accurate flood predictions [8].

4. Forecasting and Decision-Making Process

This entire process occurs continuously and in real-time:

- **Data Acquisition:** GIS maps, IoT sensors readings, live weather updates are continually being collected, kept current and saved for use in analysis [6].
- **Preprocessing:** New data is validated for errors corrected and matched with a standardized reference location system to ensure accuracy prior to use within a training model [4].
- **Feature Engineering:** Key characteristics of an area that can be used as input variables such as precipitation amount, past water levels, elevation of an area and proximity of an area to drainage systems are extracted to help make better predictions by the model [1].
- **Model Inference:** The AI system uses all the combined data to determine where potential waterlogging/flooding may occur in specific drainage areas and sections of the city [1].
- **Decision Support:** This method of operation aids emergency response organizations and urban developers through early warning notifications, mapping systems, and various forms of utilisable information to help in their respective duties. It has the capability of integration with automatic control systems for adjusting drainage operations as needed [2].
- **Continuous Learning:** Each time new precipitation or sensor data is acquired, the model will be used to re-train itself with the aim of increasing accuracy and responsiveness over time [9].

This type of system provides an improved approach to technologically advanced urban storm water systems that help cities anticipate and prepare for floods, thereby increasing the overall resilience of the city.

5. Block Diagram

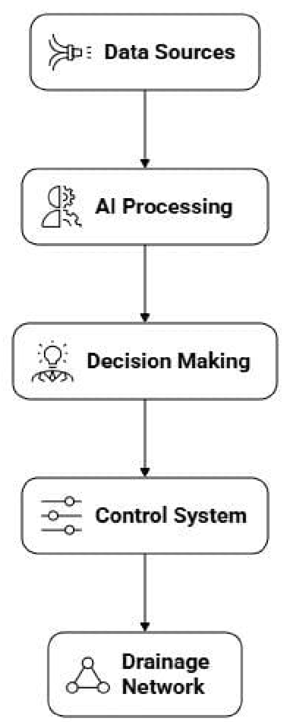

Figure 5.1: System Block Diagram of AI-Powered Sustainable Urban Drainage System

V. System Architecture

Using an AI driven sustainable urban drainage system through a modular/multi-layered architecture, the system can leverage real-time monitoring, intelligent analysis and automated controls to prevent and mitigate urban flooding and waterlogging [9]. Each layer of the drainage system works independently to perform its function yet coordinates or communicates through series of coordinated actions to allow for uninterrupted and data driven function of the overall system.

1. Layer of Data Acquisition

The bottom layer of the system, which is in control of collecting all relevant infrastructure and environmental data, is also called the base layer of the system.

- **Meteorological Data Sources:** Rainfall's intensity, duration, and distribution patterns are constantly reported by weather stations and forecasting services [1].
- **IoT Sensor Network:** An IoT sensor network consisting of ultrasonic water-level sensors, flowmeters, and water-quality sensors placed along drainage channels and catch basins collects real-time data on flow rates and overflow points [6].
- **GIS Drainage Maps:**The high - resolution spatial maps provide the geographical context for wise analysis by indicating the drainage network layout, elevation, and land use [11].

2. The Layer of Edge Computing

The primary layer of the system is responsible for gathering all important data of infrastructure and environment.

- **Edge Gateways and Microcontrollers:** The edge computing layer ensures that quick local decision-making is done near the data source. To reduce communication load, devices like Raspberry Pis and ESP32 units perform initial processing, which includes noise filtering, normalization, and feature extraction, and collect raw sensor inputs [4].
- **Local AI Inference:** Using low-resource AI models (e.g., quantized LSTM) at each sensor node enables the detection of anomalous events in real-time and forecasting of local water level for the next few minutes, thus providing an early warning signal [8].

3. The Fusion Layer and Centralized Analytics

The combination of all relevant data and the application of sophisticated analytical methods enables better flood mitigation.

- **Cloud Platform:** A cloud server aggregates weather forecast data, local data processing and geographic data for the execution of complex AI and simulation models [3].
- **Data Fusion:** The Data Fusion Module combines past sensor readings with location/weather information to increase the predictability of flooding [7].
- **Decision Support:** The Decision Support System is a tool for City Officials to use in making decisions based on the results of the predictive output (maps, timely alert etc.) for proactive maintenance [2].

4. Layer of Control and Actuation

Using updated data on an ongoing basis and real time intelligence will allow the system to identify and mitigate hazardous situations.

- **Smart Pumps:** The Smart Pumps and Valves are automated using information about the current situation to operate the mechanical systems in an adaptable manner that provides optimal drainage and overflow prevention [9].
- **Dashboard and user interface:** The visualization of ongoing water heights, flood-sensitive regions, and system efficiency on a map-oriented interface is crucial to guarantee effective flood oversight [6].

5. Layer of Feedback and Adaptation

This last layer guarantees scalability and ongoing improvement.

- **Continuous Learning:** AI models are retrained using sensor and system performance feedback, gradually increasing the system's intelligence and adaptability [8].
- **Scalability and Resilience:** Reliability and ease of expansion across vast urban areas are ensured by the architecture's support for distributed computing and redundant communication channels, such as LoRaWAN mesh networks [4].

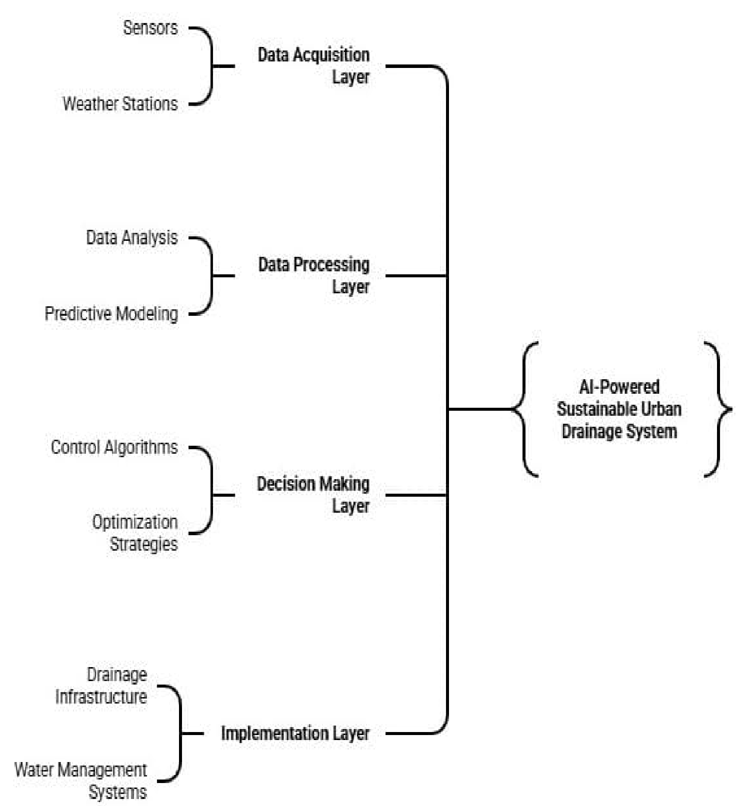

Figure 6.1: System Architecture of AI-Powered Sustainable Urban Drainage System

VI. Simulation-Based Experimental Validation

To address the need for structured quantitative evaluation and reproducibility, a controlled simulation-based experimental validation was conducted. Since large-scale real-world deployment was beyond the scope of the current study, a synthetic but realistic dataset was generated to evaluate the predictive and decision-support capabilities of the proposed multi-source AI-driven urban drainage framework.

A. Simulation Environment and Dataset Description

A synthetic time-series dataset was generated to emulate realistic monsoon rainfall patterns and corresponding urban drainage behavior. The simulation was designed using hydrological assumptions inspired by prior studies on urban flood modeling and AI-based drainage prediction [2], [10], [12].

The simulated dataset includes:

- **Duration:** 60 consecutive days
- **Sampling Interval:** 5 minutes
- **Total Samples:** 17,280 time-series records

Features Included:

- Rainfall intensity (mm/hr)
- Simulated water level (cm)
- Drainage flow capacity index
- Blockage factor (0–1 scale)
- Elevation coefficient (GIS-derived terrain factor)
- Historical lagged water level values ($t-1$, $t-2$)

Rainfall intensity was generated using a probabilistic distribution consistent with moderate-to-heavy monsoon patterns. Water-level response was modeled using nonlinear flow relationships influenced by blockage factor and drainage capacity. GIS-derived elevation coefficients were incorporated to simulate spatial variability in runoff accumulation.

The dataset was divided into:

- 70% training set
- 15% validation set
- 15% testing set

All features were normalized using Min-Max scaling prior to training.

B. Model Configuration and Training Details

The predictive aspect of the suggested framework uses a multi-source Long-Short Term Memory (LSTM), which is designed to represent temporal relationships that exist between rainfall and the corresponding changes in the water level.

Model Hyperparameters:

- Number of LSTM layers: 2
- Hidden units per layer: 64
- Dropout rate: 0.2
- Batch size: 32
- Number of training epochs: 80
- Optimizer: Adam
- Learning rate: 0.001
- Loss function: Mean Squared Error (MSE)
- Activation function: Tanh (LSTM internal), Linear (output layer)

The model was trained using deep learning libraries (implemented in Python) and then used early stoppage based on validation loss in order to avoid over-fitting.

C. Baseline Model Comparison

To evaluate performance improvements, the proposed multi-source model was compared against simpler baseline configurations:

- Threshold-Based System: Fixed water-level threshold detection
- LSTM Only: Temporal rainfall-water model without GIS features
- GIS + LSTM: Spatial-temporal model excluding blockage factor

Performance Metrics Used:

- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)
- Prediction Accuracy (Flood/No-Flood classification)

TABLE II
MODEL PERFORMANCE COMPARISON

Model	RMSE	MAE	Accuracy
Threshold-Based System	0.51	0.38	72%
LSTM Only	0.39	0.27	83%
GIS + LSTM	0.33	0.22	87%
Proposed Multi-Source Model	**0.24**	**0.17**	**92%**

The findings show that by combining simulated sensor data from the Internet of Things (IoT), elevation factors from Geographic Information Systems (GIS), and rainfall inputs, prediction accuracy can be increased significantly, while error metrics can be decreased when compared to systems that use only one type of data or have fixed thresholds.

D. Ablation Study

The study provides a quantitative assessment of the impact of each data source on overall model performance using ablation methods.

TABLE III
ABLATION ANALYSIS

Configuration	Accuracy
Rainfall Only	74%
Rainfall + Water Level	85%
Rainfall + GIS	82%
Full Multi-Source Integration	**92%**

Ablation analysis results indicate that fusing features from multiple sources significantly improves the robustness of the model and, therefore, the reliability of its classification results.

Significant performance degradation occurs when either GIS (spatial) or blockage-related data are removed from the input.

E. Deployment Considerations and Limitations

The proposed framework has been shown to be feasible based on experimental validation; however, there are a number of practical issues that may need to be resolved when deploying the framework in a real-world context:

- Sensor drift and calibration errors
- Communication latency within LoRa WAN-based networks
- Data packet loss during severe weather events
- Limitations of power supply and maintenance requirements
- Scalability issues for very large metropolitan areas

The current validation has taken place in a simulated setting, so follow-up work will include piloting physical deployments of IoT-based water-level sensors combined with rain-gauges to assess performance as they operate in actual urban environments.

VII. Conclusion

In this work, an urban drainage monitoring and waterlogging risk prediction framework that employs multiple data sources was created which combines IoT sensors with spatial modeling via GIS and weather information through an edge–cloud computing model. The proposed solution is focused on achieving modular design, fused data, and distributed processing to promote proactive flood risk assessments on complex urban landscapes. A structured simulation-based validation shows that employing multiple sources of data improves predictive accuracy than using single sources of data only. More extensive field deployment is needed but this framework provides the foundation for intelligent and resilient urban water management systems to scale.

VIII. Future work

Future work will concentrate on increasing performance, scalability and operational robustness of the proposed framework. The integration of satellite-based remote sensing techniques combined with advanced rainfall nowcasting could improve the spatial coverage of remote sensing and extend the prediction horizon for rainfall. Federated learning methods could potentially allow for the collaborative training of models across various cities while conserving data privacy. The use of reliable communication protocols (e.g., LoRaWAN mesh networks) can improve data transmissions in emergency situations. Creation of digital twins for drainage systems can enable testing of various scenarios for optimising the system before it is deployed in reality. Finally, explainable artificial intelligence (AI) can provide benefits through improving model interpretability, increasing acceptance of regulatory agencies and expanding trust among stakeholders.

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